

Mobile network design and optimisation

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The rapid design and initial deployment of cellular networks is now commonplace throughout the world. Once launched, explosive growth in the customer base is the norm — in the UK, for example, one million new customers are being added every 6 months, out-stripping the rate of fixed network connections. This presents the very significant challenge of how to rapidly roll out networks with the desired quality and capacity, yet at minimum cost. In this paper skills, tools and techniques are presented that BT Laboratories has developed and exploited to support the growth of the Cellnet analogue (TACS) and digital (GSM) cellular networks and the initial design of GSM networks for overseas licence bids.

1. Introduction

Cellular telephony has established itself as the major success in telecommunications over the last ten years. Startling growth rates and tumbling equipment costs, brought about by international standardisation and competition, characterise all these networks world-wide. In the Nordic countries, which have been in the forefront of this revolution, penetration rates of 25% are common and fixed telephony replacement is now established. Also, the amount of traffic and type of information being carried by cellular systems is changing. As customers become accustomed to voice communication ‘to any one, anywhere, at any time’, so they are now expecting the same for data communications. With the increasing power and shrinking size of notebook personal computers, the ‘office on the move’ concept is becoming reality, increasing the amount of data traffic being presented to the network.

BT has played its part in this revolution through its 60% share holding in TSCR (Cellnet), its sales provision channel through BT Mobile and in overseas cellular licence bids and operations. Cellnet has been operational now for over ten years and has built well over 1000 base sites to support its analogue (TACS) and digital (GSM) networks. BT Laboratories (BTL) has supported Cellnet both in the design and expansion of its networks and have provided technical support for overseas licence bids and operations. To undertake this work successfully, skills and toolsets have been developed to meet the stringent market needs. But what are the engineering problems?

A cellular network contains nearly all the main components of BT’s fixed network with the addition of a large radio network. The challenges that a mobile radio network presents relate not only to the performance of radio links (a non-trivial problem with error rates often worse than 1 in 10) but also to the additional network loading necessary to support mobility, with capacity being dimensioned statistically throughout the network. This is one of the major drivers for the engineers who try to provide the optimum quality at minimum cost, heightened further by the tough competition that is met in nearly all such networks — Cellnet are currently facing three competitors who are all fully committed to winning market share. This strong competitive driver for cellular gives rise to another characteristic — high annual churn rates (about 25% in the UK).

To compound the drivers already mentioned, the cellular engineer has to address the additional problem that the radio resource is strictly limited — for example on the Cellnet TACS network nearly 2 million subscribers are supported on just 600 radio channels. Frequency planning and reuse are key problems.

The problem for the cellular engineer is thus complex — to deliver a quality product to the customer (wherever he might be) at minimum cost within the tight constraints of the spectrum available. This paper traces the development of a cellular network from the design of a new network ‘green field design’, through planning and implementation

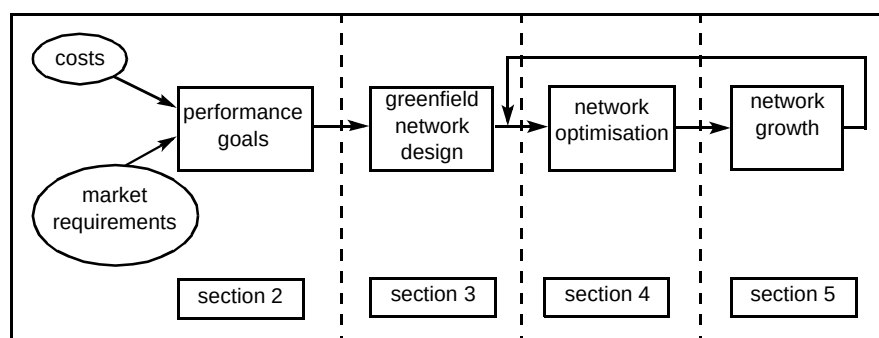


Fig 1 Life cycle of a cellular network.

to growth and replan. The technical challenges met, and solutions employed, by BTL are described.

The life cycle of a cellular network is depicted in Fig 1, with the relevant section of this paper also indicated.

2. Mobile network design performance goals

2.1 Quality of service factors

For customers, the expectation is that they should be able to ‘make and reliably receive uninterrupted calls and converse with ease’; clearly the cost to the mobile operator of providing this level of service in every conceivable location and at any time would not result in a viable business.

Realistic quality of service (QoS) performance goals have therefore to be in place before commencing the mobile network design. At the highest level this splits into radio coverage, availability, and grade of service (GoS), but many other goals need to be defined, such as those to do with services, billing, operational and network management requirements. It is also arguable that customer-facing service is part of the QoS experienced by customers; however, this is outside the scope of this paper.

The GoS offered to customers is directly related to the capacity of a network. Careful analysis of existing and predicted mobile traffic patterns, i.e. where it is generated, has to be made to ensure that capacity is available at the right locations. Where no existing cellular network operates, it is the job of marketing people to predict the likely levels of traffic and pass this information on in a format suitable for radio and network designers. Unlike a fixed network, mobile customers are not always predictable in their movements and traffic overloads can occur when abnormal events take place, such as bad road conditions or county shows.

The availability of the network is defined as the percentage of time the network is in working order. It considers all the network components from the base station

antenna to the point of interconnect to the public land network. It does not consider the percentage of land ‘covered’ by the network. The availability of the network is directly related to the reliability of network components and service restoration time after a fault appears. Mobile networks are complex and it is necessary to pay particular attention to reliability of the network components and how quickly they can be repaired. This latter point is very important in the design of mobile networks for undeveloped countries where journey time may be significant compared to the UK, or where access to towers can only be made in the day time.

Finally, coverage has to be sufficient to ‘capture’ traffic and provide apparent ubiquitous service to customers. Providing coverage to places where no one goes is clearly not cost effective.

The expected quality of a call, which is determined both by the integrity of the radio link and transmission plan required to deliver a good quality call, is also covered under this heading. The GSM system adds another dimension to this, because of its data handling and messaging capabilities. The system needs to be dimensioned such that all information, be it speech or data, can be carried by the system with an acceptable GoS. The cellular designer, therefore, needs to consider the impact of not only voice communication requirements but also the possibly more stringent requirements for data communications.

2.2 Setting goals

A useful starting point is to establish the balance required between the performance to be enjoyed by customers using the network and how much the mobile operator is prepared to invest.

A compromise has therefore to be made, between a high probability of being able to make a good-quality call in the majority of locations, and the cost of providing a network. Any savings which can be made through efficient use of radio spectrum and network infrastructure is therefore a

bonus. The design of efficient mobile networks is the main subject of this paper.

A network must be designed to provide good QoS through network 'roll-out' and rapid expansion. The majority of cellular operators have experienced a very high growth of customers in the past decade. This growth has sometimes been difficult to predict and it has been even more difficult to ensure that QoS is maintained at the desired level.

The challenge through network expansion is to increase coverage and capacity while maintaining or improving quality. This is generally met by adding new cells, reducing the size of, or splitting, existing cells and acquiring new radio channels (if available). Careful consideration of 'up and down' link budgets (base-to-mobile and mobile-to-base signal loss) and the traffic to be handled by the new cells is needed to ensure that quality is not degraded. This relationship is shown in Fig 2.

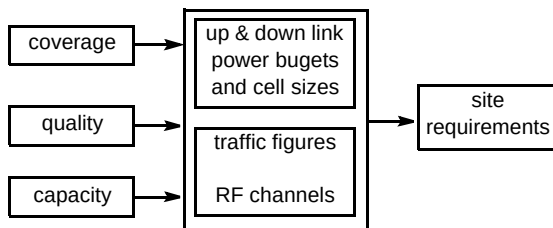


Fig 2 Roll-out challenge.

2.3 Typical performance goals

It is useful to consider typical QoS performance goals for a mobile system. The probability of a poor connection over a mobile radio link is more likely than on a wire-line telephone, because of the cost of providing an absolute, ubiquitous service. The goals are therefore split into radio and fixed parts, but the overall probability of not obtaining service is a product of the two parts. It should be noted that the following network design performance goals are generic and only illustrative.

- Radio coverage
In-building coverage in urban and suburban areas, coverage for all conurbations of over 1000 population, industrial estates, railway stations, motorways and 'A' roads, with less than 10% outage probability.
- Transmission quality
The transmission quality is governed by a number of parameters, many of which are purely determined within the mobile terminal. They include delay, quantisation distortion, loudness ratings, echo and echo control and other terminal acoustics. ITU-T

recommendations can be used, together with GSM standards, to formulate a transmission plan for the network designers and interconnect negotiators. Transmission issues are explored in more detail in section 3.4.

- Network grade of service

Overall blocking probability of 3% on a direct mobile to point of interconnect (POI) connection (due to 2% blocking in mobile access to base stations), 0.5% on base station to mobile switching centre (MSC) links and 0.5% on MSC to POI links.

- Network availability

Greater than 99.9% overall.

2.4 Quality of service measures

A customer only has access to a mobile terminal and a service provider, who may, or may not, also be the radio network operator. The customer is the only accurate point at which QoS measures can be made; however, the network operator has access to event and statistical performance data from the network, which is valuable for assessing overall performance.

QoS taken from the customer 'viewpoint' can be assessed by the following measures:

- blocking probability,
- successful / dropped calls,
- call set-up and clear-down times,
- opinion of call quality (audio and data),
- radio coverage / ubiquitous service,
- customer feedback and billing complaints.

Mobile operators expend significant effort in measuring the performance of their network in ways to imitate the customer 'viewpoint'. This led to the development of automatic equipment capable of making calls from mobile to fixed networks and vice versa.

The mobile network provides a multitude of performance measurement points which can be used to build a picture of QoS provided to the mass of customers. The measures from these points are also closely related to the network dimensioning and performance criteria which were defined during the network design process. This means that the design performance of a network may be validated through analysis of network-based measurements.

The following measures are typical of those available:

- switching nodes — processor loads, congestion, holding times, queue lengths, etc,
- signalling network — usage, link and channel status,
- trunking network — busy-hour call attempts, answer-to-bid ratio, answer-to-seizure ratio, etc,
- radio network — coverage, interference, dropped calls, etc,
- handover performance — success rates, failure reasons, counts, etc,
- global performance — total traffic, call completions, number of customers, etc.

The monitoring of these measures has become more sophisticated over the years, with collection being performed centrally at operations and maintenance centres (OMCs). Work on better interpretation of network-collected measures, by the use of visualisation and trend analysis, is ongoing in an effort to make them more useful to radio and network planners.

3. Greenfield design

The design of a mobile network is best divided into two sections — the radio network consisting of the cell sites and the mobiles, and the fixed network extending from the cell sites to the interconnections to the public land networks. Also, the transmission aspects of the fixed network are complex and need special consideration. These aspects of designing a new mobile network are described in detail in the following sections.

3.1 Greenfield radio network design

New GSM networks are now being conceived, planned and built within progressively shorter time-scales. It is, therefore, critical that radio network planners can rapidly design the basic network of radio base stations which will form the launch network. In particular, the initial network must provide adequate capacity and coverage, have minimal radio interference problems, and provide good call quality. Additionally, the design of the network should allow coverage and capacity to be increased progressively as the number of customers grows. This requires that the radio planners understand how their initial network will form the foundation for the mature network of the future.

To aid the radio planner in this task, BTL have developed a software tool called 'Auto'. Auto is an automatic BTS placement tool, which is able rapidly to design greenfield cellular radio networks to meet specific coverage or capacity criteria. In addition, Auto can be used

to estimate the number of additional base stations that will be required in a network as the coverage and capacity requirements of the network grows.

In order for Auto to generate practical base-station networks, it must be supplied with information about the basic morphology of the area to be covered by the new network and the capacity and coverage which can be achieved with various base-station designs. The morphology of the area is generally divided into a small number of main classes such as dense urban, urban, suburban, rural, water, etc. This enables the algorithm developed in Auto to assign a proportion of the total offered traffic to the individual morphology classes to produce a network traffic map. Dense urban areas, typically the centres of large cities with their high concentrations of business, will be assigned the highest levels of offered traffic while water areas can be considered as having negligible levels of offered traffic.

It is also important for the radio planner to ensure that the network has balanced radio links. A cellular radio channel consists of two radio links, the uplink and the downlink. The downlink is the radio path from the base station transmitter to the mobile telephone receiver, and the uplink is the corresponding radio path from the telephone transmitter to the base-station receiver. The maximum range of a mobile telephone is determined by when either the base-station receiver or the mobile-unit receiver can receive an acceptable signal to ensure the call quality of the corresponding link. In designing a radio system with balanced links, the received signals will become unacceptable simultaneously, with both the uplink and the downlink call quality degrading. This is the ideal situation, since the mobile telephone user will be experiencing similar call quality degradation to that experienced by the person they are speaking to. This is what users intuitively expect when a mobile telephone call begins to fail.

The base-station design information, particularly the maximum range, is used to enable the algorithm to position base stations such that the design philosophy of the launch network is achieved. If it is important for the launch network to achieve extensive coverage, usually major conurbations and transport networks, the network will normally be formed from large sectored base stations. These base stations have high transmit powers, and by using directional antennas to form each sector they can achieve coverage over a wide area. Alternatively, the initial network may be aimed at achieving good coverage purely within the main towns and cities, particularly within buildings. In this network, the coverage will be better achieved with many small, lower power base stations. Although the coverage area of these base stations is small, the higher density of base stations ensures that the signal levels that are achieved within buildings is above that which would be achieved with a single high-power base station.

Once Auto has been supplied with the basic morphology, and hence traffic information, and with the types of base stations which will be used to meet the network design criteria, it is able to design the basic radio network. In a greenfield design, Auto is not constrained by the coverage offered by an existing network. Therefore, it uses a very simple algorithm to place base stations into the new network. The area of interest is initially covered by a regular grid of base stations. The type of base station placed at any location will depend upon the morphology class at that location. Auto will calculate how much of the offered traffic will be carried by each base station. It then enters an iterative optimisation process removing base sites until all the remaining sites are required in order to meet the network GoS requirements identified in section 2. This process is depicted in Fig 3. The map on the left of Fig 3 shows the initial regular grid, and the map on the right shows typical output when Auto has optimised the base site locations. The non-uniformity in the right hand map is the demand caused by clustering traffic in and around towns and cities.

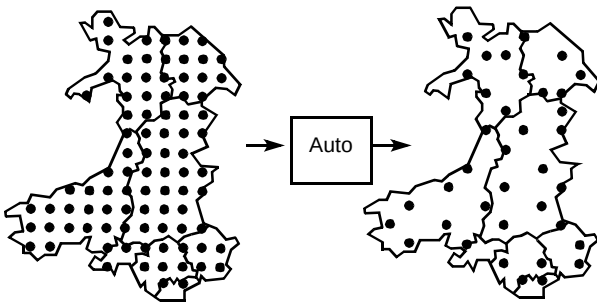


Fig 3 Process of automatic base-station location optimisation used by Auto.

The output of Auto cannot be used directly to build a new cellular network because the proposed base-station locations may not be achievable in practice. The actual position of a base station depends upon a suitable location being identified. The location must enable adequate coverage of the target area to be achieved, be easily accessible for maintenance, and planning permission may need to be obtained. In many cases, there may even be preferred locations for base stations, such as buildings that are owned by the companies involved in the formation of the new cellular radio telephone system. The new operator will need to start site acquisition processes in the vicinity of the locations suggested by the tool. When potential sites have been identified and legal access achieved, detailed radio coverage calculations can be made for the exact location, antenna height, local obstacles, etc. In some cases it may be necessary to perform a 'drive' survey to confirm acceptable coverage can be achieved from the potential site.

Once the base station positions have been finalised, the radio planner must optimise the base site's transmit power, to give optimum coverage while causing minimum interference, and assign radio channels to each base station.

Unfortunately, real cellular radio networks seldom approach the ideal regular cell patterns used in theoretical frequency reuse calculations. These constraints can result in inefficient radio-channel allocation in order to achieve acceptable network performance. To aid this process, radio planning tools are used to give accurate radio propagation data. These tools also normally have the ability to perform automatic frequency planning (AFP) to optimise the radio channel assignment with minimal intervention from the radio planner. Radio planning tools are described in more detail in section 5.1.4.

3.2 Fixed network design

The design of the fixed side of a mobile network consists of the interconnection of the cell sites to the public land network. Although all types of mobile network have this requirement, this section will consider the European digital cellular system, GSM, as an example. A GSM fixed network consists of three main components — base transceiver stations (BTS), base station controllers (BSC), and mobile switching centres (MSC). BTSs provide the air interface to mobile users and are the parts which form the cell sites. These are controlled by BSCs, which in turn are parented on to the MSCs. The top layer of a GSM network is the interconnection of MSCs and the public land networks (PLN). Figure 4 is a schematic design of a GSM network.

The fixed network designer has to work out the minimum cost interconnect strategy for the GSM system components in order to meet the overall GoS requirements of section 2. BTSs, by necessity, have to be located remotely from the MSC site. BSCs can, however, be located either remotely, usually to optimise transmission costs, or at an MSC site. Interconnects between MSCs and the PLN can also generally reduce the mobile operator's costs if the majority of the transmission is contained within the mobile operator's network, using the PLN only for the final connection to the local loop.

An ideal design process would be able to design the whole network from the parenting of BTSs, to the interconnection of MSCs and the PLN. However, due to the number of different possible topologies available, along with the constraints imposed by equipment/site limitations, the computational time required for an 'ideal' minimum cost network is prohibitive. From experience, network design is therefore split into two — optimisation of the access network, defined as from an MSC to the BTS, and optimisation of the backbone network, defined as the interconnection of the MSCs and PLNs.

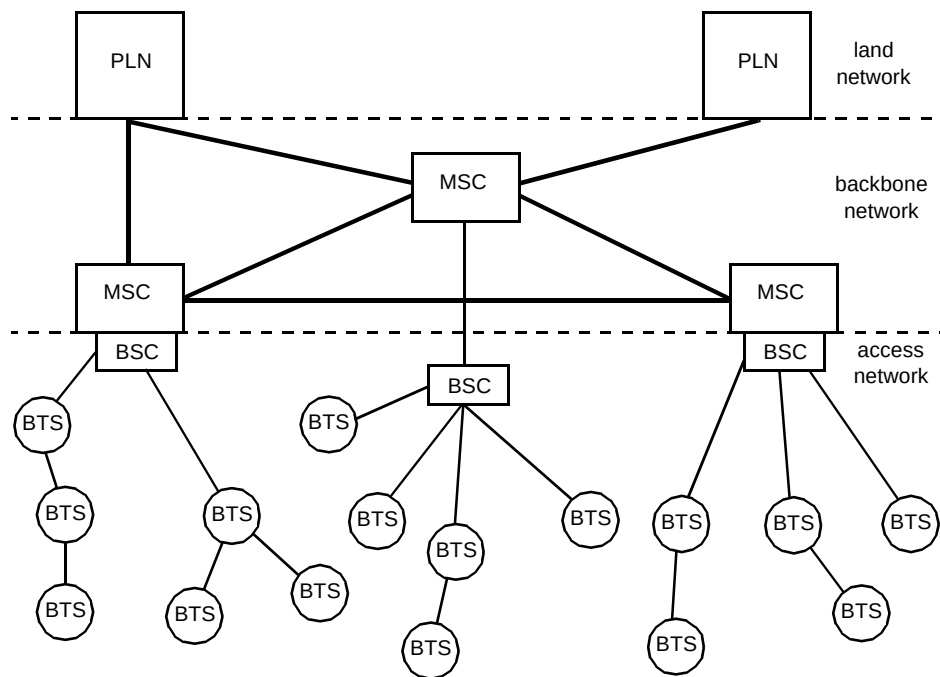


Fig 4 Schematic design of a GSM network.

Before any design work can commence there are a number of key design decisions to be made:

- where and how to design the architecture to deliver efficiently such services as voice messaging, short messaging service (SMS), etc,
- where to place the MSCs (affects backbone and access network design),
- how to interconnect MSCs and land networks to maximise revenue (affects backbone design),
- how to parent BTSs to MSC sites cost effectively (access network design).

3.2.1 Backbone optimisation.

For any greenfield network a set of preferred MSC sites would be known. This is usually derived from the analysis of demographic data or from other constraints such as the proximity of suitably PLN interconnect points, or the availability of suitably sized and equipped accommodation. Additionally, from pre-analysis, it has to be determined whether the access or backbone network is the main cost driver of the final price of the network.

To optimise the interconnection of MSCs and the land network within a greenfield design, it is important to know, or deduce from marketing data, how customers will use the network. Data is needed, on a regional basis, for the daily traffic profile of users, the proportion of land to mobile and land to mobile calls, the proportion of non-voice data, the

proportion of local, national and international calls, call-holding time profiles, etc. Each of the above questions are important as they allow calculation of the daily revenue that will be generated for different network configurations or MSC placement. For optimisation of the backbone network it is usually assumed that within the access network all BTSs are parented to their closest MSC site, this being the simplest network topology to implement.

There is no easy process for backbone network optimisation — all possible configurations are usually analysed, for a number of different years. Best placement and roll-out of MSC sites, best PLN interconnect points, and whether near end or far-end handoff is to be implemented can all be decided from the analysis.

3.2.2 Access optimisation

The simplest topology for an access network is direct star. This means that all the BTSs are directly connected via leased line bearers (megastreams, etc) and/or other carriers, such as microwave, to MSC sites. Here BSCs are co-located at the MSC site. However, because of the pricing structure of leased line bearers, there is usually scope for reducing the cost of the access network significantly. The most common types of access network topologies are:

- remote star — a subset of BTSs are parented to a remote BSC site or multiplexer before being routed to the MSC site,

- link-multidrop — BTSs are chained back to the MSC site,
- resilient rings — similar to link-multidrop but the tail end has a return path to the MSC site,
- combinations of the above.

Detailed descriptions of network topologies are described in section 3.3.

3.2.3 Access network design tool.

Over a number of years BTL has been developing an access network design tool — the mobile network design tool (MNDT). The aim of MNDT is to minimise the cost of connecting the elements of a fixed network using leased lines. Additional costs such as the cost of multiplexers, site preparation costs, switch costs, connection charges, annual maintenance charges and equipment depreciation are all taken into account. The network is configured to comply with the GoS criteria defined in section 2.

Early versions of MNDT relied on clustering techniques for optimisation. Later versions of MNDT utilised algorithms from minimum spanning trees to non-hierarchical clustering. The latest version of MNDT relies on a simulated annealing algorithm. With this algorithm additional cost savings of 10-20% are possible against the

earlier versions. It also has the capability of designing a large range of different topologies.

3.3 Transmission design

The transmission network is effectively the skeleton of a mobile network. The operator hangs the services and platforms from it, forming the body of the network. Currently, most mobile networks within the United Kingdom and the world use relatively traditional transmission networks to connect the cell sites to the switching centres. Leased 2 Mbit/s (E1) circuits are used to make the connections. Where licences allow, microwave circuits are used, particularly for the final hop to the cell site. The frequencies most used include 38 GHz and 23 GHz bands. The reason for this basic design is the speed of network roll-out required by the mobile operators in a rapidly expanding market-place and the restrictions placed upon them in terms of infrastructure provision by the local regulator. With many operators now moving into a more mature market where reduced tariffs, often matching those of the PLN, are required, the need to re-examine the transmission network to reduce operating costs is obvious. In addition, the telecommunications liberalisation policy within the European Union (EU), is allowing mobile operators much greater flexibility in infrastructure provision before similar flexibility is offered to new fixed network operators. New transmission technologies and network topologies offer the network operator the opportunity to

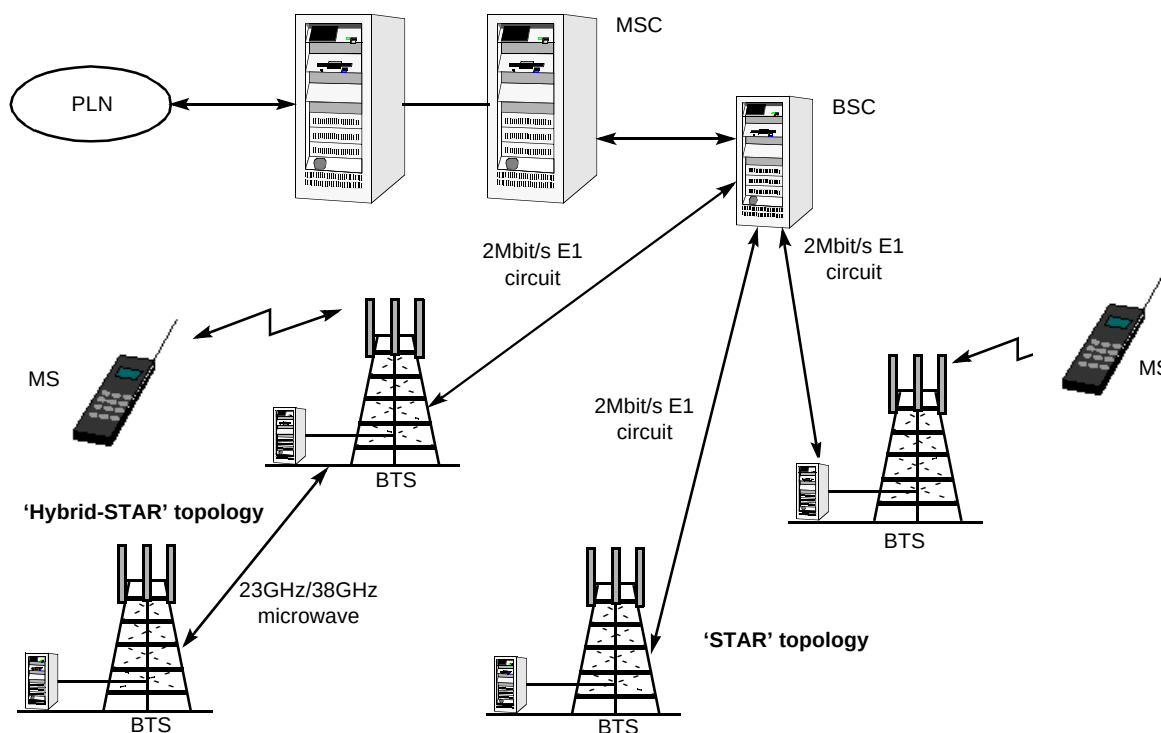


Fig 5 Standard star and hybrid star transmission network topology.

reduce operating costs. As new network topologies and technologies are implemented, the need for effective network synchronisation increases in importance.

3.3.1 Transmission technologies and topologies.

As has already been stated, most mobile network operators currently use either a standard star network topology with 2 Mbit/s E1 circuits connected directly from each cell site back to the switch site, as illustrated in Fig 5.

The main issue with a standard star transmission network is that many of the 2 Mbit/s E1 circuits are under-utilised, which is inefficient for the network operator. While a typical two-carrier GSM cell requires 320 kbit/s of transmission bandwidth, a large cell of six carriers requires only about 1 Mbit/s of transmission bandwidth. By adopting a different transmission network topology, such as a hybrid star, where a combination of 2 Mbit/s circuits and microwave circuits are used, the network operator can make more efficient use of the transmission bandwidth available

and create opportunities for using new transmission technologies.

The resilience can be improved, if required, by using diverse path techniques and back-up paths. If the BTS to BSC path is a microwave circuit then $1 + N$ back-up protection can be employed to protect the traffic-carrying microwave connection. In addition, a diverse parallel path can be provided by either a cable connection or a microwave mesh network, or indeed a combination of the these techniques. Figure 6 illustrates the diverse path and mesh techniques.

By positioning a point of presence closer to the cell sites than the BSC, such as a larger BTS site, the network operator can utilise new transmission technologies to connect to the cell site. Recent developments in high-speed modem technology enable the use of high-speed digital subscriber loop (HDSL) technology. This allows up to 2 Mbit/s of bandwidth to be obtained over a number of copper pairs, between two and four, over distances of up to

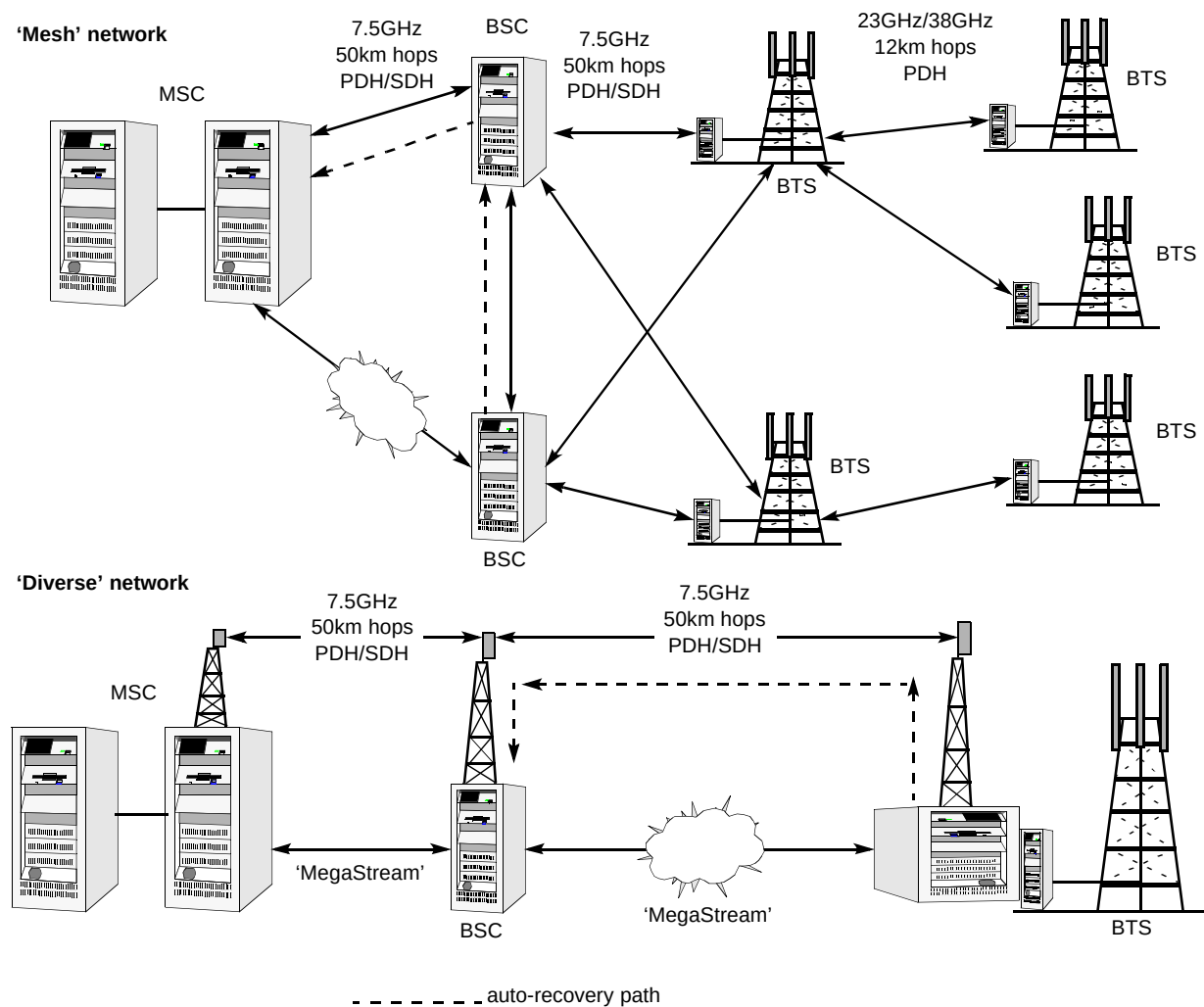


Fig 6 Diverse and mesh networks for improved resilience.

5 km, dependent on the gauge of the copper and the bit rate required. This technology is ideal for deploying an urban microcellular network or for connecting in-building systems. The technology is reliant on the cellular network operator being able to obtain copper pairs from an incumbent local loop operator. As the local loop evolves from a copper-based network to a fibre information super-highway, the cellular operator could evolve the urban HDSL topology to utilise a passive optical network (PON) technology.

Another new technology that can be used to connect the BTS and BSC sites is the integrated services digital network (ISDN). Cellular equipment manufacturers are currently developing ISDN interfaces for their equipment. The ISDN would not be the primary transmission media for the BTS site because of the call-charging regime over a twenty four hour period. However, many BTS sites have one or two traffic peaks throughout the day. If the BTS's permanent bandwidth is dimensioned according to the average daily traffic, the ISDN can be used to supply sufficient bandwidth for the traffic peaks, saving transmission costs. In addition, the provision of temporary cell sites for sporting events and conferences can be achieved via the ISDN. Figure 7 illustrates how the ISDN can be used.

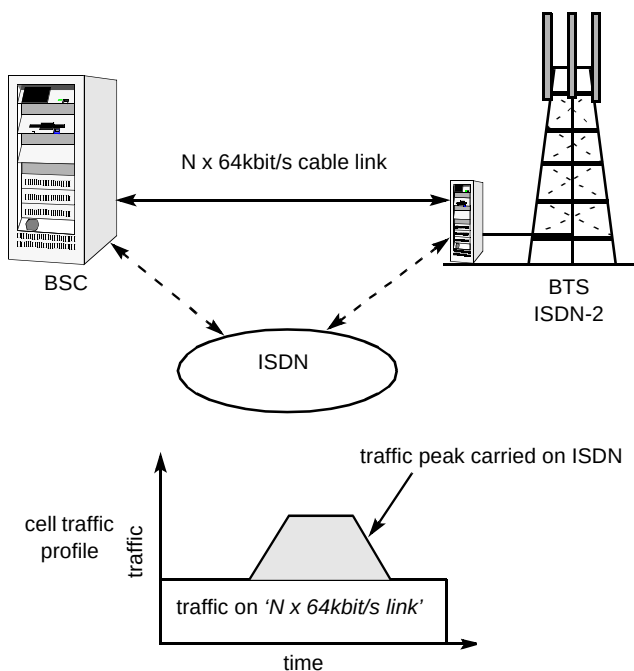


Fig 7 Use of ISDN for cellular transmission provisioning.

So far, the transmission network has only involved the used of traditional pleisiochronous digital hierarchy (PDH) technology. However, synchronous digital hierarchy (SDH) equipment is becoming increasingly available and can be of use to the cellular operator. The basic SDH transmission rate is the STM-1 155 Mbit/s. For individual cells this is excessive and even using some form of 'ring' topology would be difficult to justify unless several hundred cells

were connected in the ring. However, SDH can be used in the BSC-to-MSC and MSC-to-MSC network, using resilient self-healing rings to replace fully interconnected PDH mesh networks, this reduces the amount of equipment required at these sites and the amount of cabling.

Although most cellular network operators currently use the standard star or the hybrid star transmission network topology there is considerable scope for producing more cost-effective, resilient and manageable transmission networks by adopting new topologies and technologies to connect up the cellular network components.

3.3.2 Network synchronisation

Many cellular network operators currently draw their synchronisation source from the incumbent operator's national master clock. The synchronisation hierarchy is normally distributed on a master/slave basis, the cellular operator drawing synchronisation from the trunk switches with which it interconnects for delivering traffic from the mobile network to the PLN. The synchronisation is then disseminated through the cellular network on a master/slave basis from the MSC down to the BTS via the BSC.

As the cellular operator interconnects to other network operators, who may or may not use the incumbent's synchronisation source and may connect to the incumbent at a different level in the synchronisation hierarchy, there is a need to re-examine the design of the synchronisation network. In addition, the difference in synchronisation between a cellular operator's self-provided microwave and a leased 2 Mbit/s circuit could be sufficient to cause slips on the network and hence reduce the quality of service offered to the customer.

Probably the best method to remedy the situation is for the cellular operator to install their own synchronisation source. Until recently this would have involved the purchase of prohibitively expensive caesium atomic clocks with a lifetime of three to five years. It is now possible to draw accurate synchronisation directly from the global positioning system (GPS) network of navigation satellites.

The GPS synchronisation signal can be used to calibrate rubidium clocks which act as back-up should the GPS signal fail or the GPS receiver equipment at a given site fail. There is some concern within the telecommunications community that the GPS system can, in theory, be switched off by the U.S. Department of Defense at any time. While this is true it is highly unlikely, as the commercial use of GPS has risen by such an extent that, if GPS were switched off, the effect on commerce and industry would far outweigh any strategic gains in preventing an enemy using the system.

By using GPS, accurate synchronisation can be distributed cheaply to all points of a cellular network and, should any GPS receiver equipment fail, there is still the option to use derived synchronisation from a cellular point of presence higher in the network hierarchy.

4. Optimisation

The operator's prime concern is to maintain the system to provide the best possible service for the customers and to maximise the revenue generated by the network. To do this they must perform on-going monitoring and maintenance. This may highlight weaknesses in the original design or unexpected changes in network demand. To solve this in the short term, the operator may be able to make fine adjustments to the radio or fixed network to meet the changing needs of the growing customer base. Longer term changes in demand for the network may require network expansion which will be covered later.

Optimisation can be separated into two sections, radio optimisation and fixed network optimisation. It is likely that more effort will be needed on radio optimisation because radio dimensioning is sometimes considered an art form rather than an exact science, as is the case with the fixed network. Radio waves propagate differently in every direction, ducted by valleys or blocked by hills. Predicting possible co-channel interference caused by reusing frequencies too often is difficult — hence the need for optimisation. On the fixed side, optimisation consists more of matching capacity of the fixed network to where the customers are actually making the most calls and ensuring that optimum points of interconnect exist between different networks.

4.1 Radio optimisation

The data which allows radio optimisation functions to be performed are collected from two separate sources, network statistics which are gathered from the operations and maintenance centre and field measurements from drive testing. The OMC allows the operator to look at long-term statistics and trends for planning the growth and future capacity of the network. It also allows the operator to identify potential problems, e.g. one cell may have excessively high failed handover rates. Drive testing, however, is the only way the operator can see what the user actually experiences and is used not just for performance evaluation but also for dealing with customer complaints and questions. Drive testing is expensive, and is therefore limited to areas where problems are known to exist, as indicated by the OMC output and customer feedback. This section will review the drive test systems and how they can be used with centralised tools to improve the efficiency of the drive teams and to gain an extra insight into the performance of the cellular network.

The drive testing systems which are commercially available are from specialist manufacturers and although the individual specification of such systems vary, the basic components of the systems are similar. The main part is an intelligent mobile, which can perform as a normal mobile telephone but which also reports the engineering information used in the control of the radio link. This is available either on a dedicated proprietary interface or on a standard RS-232 port. The most commonly used phone for drive testing GSM networks is the Orbitel 900 series.

In addition, to make the data useful when reviewed, some form of positioning information is also collected. The options here include GPS, DataTrak and other dead-reckoning systems attached to the vehicles gathering the data.

The output of the phone and the positioning system are collated by a logging computer, usually a powerful notebook as indicated in Fig 8. The complexity of the software employed is what really differentiates between the systems. Systems vary in the degree of flexibility in what is collected — the amount of decoding which is carried out on the data and the complexity of the graphical or text displayed to the user vary for each system.

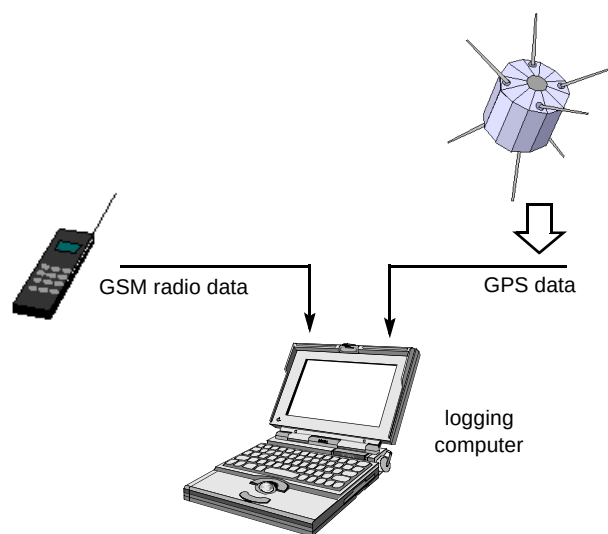


Fig 8 Typical drive data logging system.

A system of this type only gives a limited view of the system which, while good for isolating faulty equipment or other localised problems, requires the engineer to interpret the data and the interaction with the surrounding network. It has been found, in developing networks, that this is not the most efficient way of optimising or fault finding as the engineers need to see the drive data in relation to the rest of the network.

To enable this visualisation, the geographical interactive aid to network optimisation (GIANT) software suite has

been developed for use by BTL and Cellnet, to assist in the planning, fault-finding and quality improvement processes.

GIANT uses at its core an Oracle 7 database which is populated from Cellnet's radio planning tool and parameter database on a regular basis. The information loaded into the database includes the complete configuration of planned and operational cells. Information such as the location of the cell, the number of sectors the cell uses and the channels assigned to each of the sectors is loaded, along with the transmit powers and antenna types for each sector. GIANT also processes the propagation prediction information from the planning tools to provide a series of maps which show the composite coverage of the network. GIANT displays the overall coverage levels, the best server at a particular location and the worst channel carrier to interference ratio (C/I). The user can then select any of these as either a coloured overlay or as a series of line contours.

These composite maps can be placed over a graphical interface system (GIS) which holds copies of the local large-scale maps and allows the user to look at the position of the base sites, their coverage or other information as coloured overlays. Thus, the operators have a visual way of looking at the configuration of their network in real time.

By combining the output of the drive data collection tools with this view of the network and overlaying the location of the mobile, the base station to which it is operating and the channels it is using, the operator can make better judgements about the performance of the network. GIANT also displays other information from the test mobile such as the signal level received, the quality of the link and, where applicable, the transmit power of the mobile.

Typical checks which could be performed include ensuring that the handovers the mobile has actually performed match up with those predicted by the planning tool; this makes sure that the cells on the ground are the same size and shape as planned and that the interference levels are neither excessive nor distorting the shape of the surrounding cells. Of particular interest to the operator are those handovers or other events which have not occurred as predicted; these can then be fully investigated.

Figure 9 is taken from GIANT and shows an example of the main display of the drive data. Other windows allow the user to control the speed of the mobile on the screen and indicate in graphical and text form information at the point on the route. This route is shown by the black edged line which is colour coded to indicate the best server to which the mobile actually operated. The colours of the background of the map indicate the best serving sector for each point and are based on the primary channel groups in use on each sector. The location of the mobile is shown by the dark cross and the sector to which it is operating is shown shaded and is connected by a solid line. The display is updated each

time an item of location data occurs in the data file or other event occurs. The speed of replay can be adjusted by the operator. By watching the sector to which the mobile operates and the colours of the route with respect to the backdrop the handover performance can be evaluated. Other problems can be identified quickly such as cell dragging, which is shown by the green portion of the route to the south of the mobile. In this area the mobile is operating to the cell to the north of the map and will be causing interference to the cells around it. This could be due to abnormal radio propagation caused by the local terrain and it is here that the local knowledge of the engineers can make a big difference

Cellnet use GIANT as a means of checking large amounts of drive data and determining whether there are problems with the configuration of the network. GIANT also allows evaluation of improvements and changes which are made as part of the continual development and growth of the system. QoS teams continually drive routes around the country collecting drive data, which is then analysed by GIANT to highlight any problems encountered.

At BTL the system is used as a testing ground for advanced network optimisation algorithms to aid in the improvement of capacity and quality. Such algorithms can then be developed and tested on a model of a real customer network in GIANT and the benefits can be demonstrated to Cellnet before moving on to a live network. Use of GIANT for network optimisation has provided valuable lessons in the continuing development and improvements of cellular networks.

4.2 Network optimisation

Although networks are designed to cater for initial and future traffic demand, the introduction of new customer tariff packages, changes in the cost of leased circuits or changes to other licensed operators' (OLO) interconnect charges can impact on the original design. The network must therefore be continually optimised to keep costs to a minimum. Planning an optimum network is a complex task. The topology of the network, routeing strategies, cell parenting and signalling issues must all be considered. This section focuses on how the topology, interconnect strategy and routeing can be used to reduce call costs through network optimisation.

4.2.1 Topology

The topology of the majority of the fixed network is determined during the design phase and cannot easily be optimised. It is expensive to re-locate MSCs or BSCs and therefore optimisation is limited to changing the points of interconnect and the dynamics of call routeing over the

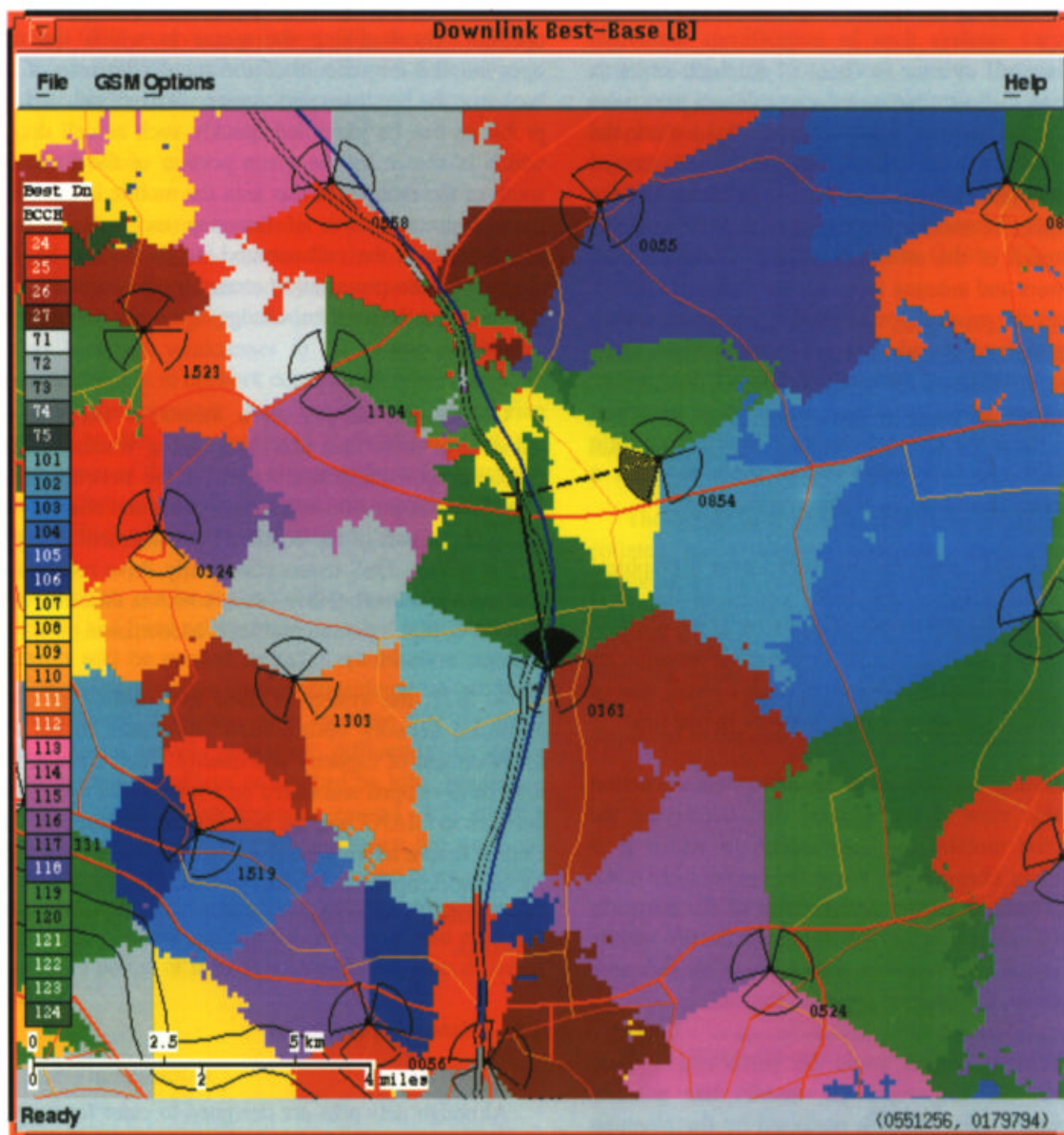


Fig 9 Typical operator's display of a drive route replay.

network, primarily between MSCs and interconnection to OLOs.

In principle, MSC interconnect can be achieved in a number of ways. Fully interconnecting MSCs will reduce the number of switch hops to connect a call and minimise processor loading. However, such a network will result in a high number of inter-switch links, a number of which will be small and inefficient. Use of tandeming will concentrate traffic over larger more efficient routes but will increase switch processor loading. Adopting a hierarchical network topology will relieve mobile switch processing and concentrate link traffic over larger more efficient routes. Such a network is ideal for the mature cellular operator environment as it allows quick and easy implementation of new services.

A hierarchical network may consist of a 'conventional' switching top layer acting as a gateway between OLOs and the operators mobile network. This top layer does not need the expensive 'mobility' features of the MSC. The lower network layer interfaces with the mobile customer via the radio network; this is used to perform all the mobility aspects of the network.

Further use of the top tier can be made to connect customers to services such as Voicebank. Unsuccessful incoming calls can be routed to recorded announcements or calls forwarded through the top layer, reducing cellular switch activity. The hierarchical top layer therefore becomes an efficient mechanism for transporting traffic.

4.2.2 Other operator interconnect strategy

If calls terminate on another operator's network, cost savings can be made by optimising the point of interconnect. The connecting cost may be significantly reduced through lower interconnect charges if calls are deposited to their nearest geographical area serving the terminating subscriber. However, the savings in interconnect charges must be balanced against the cost of carrying the traffic across the mobile operator's network. By evaluating outgoing traffic profiles and OLOs' current and future pricing strategies, it is possible to identify which traffic streams will benefit from reduced interconnect charges.

A hierarchical network will help reduce the cost of transporting this outgoing traffic. Use of a top layer of non-cellular switches to route outgoing calls from the operator's network to the cheapest interconnect point is therefore achieved without tandeming through the more expensive cellular switches. Reciprocal incoming interconnect routes can be provided at each top-layer switch, minimising the charges made by the OLOs for delivering calls to the mobile network. Intelligent routing across this top layer will ensure optimum entry to the appropriate cellular network.

4.2.3 Routing strategy

An efficient routing strategy can offer the mobile operator significant savings on the cost of interconnections between switches. The inter-switch links must be sensibly dimensioned to ensure a GoS which achieves the requirements detailed in section 2, while minimising the cost.

Simplistic routing will ensure end-to-end connection, providing adequate link resources are available. Should a link in the selected route be busy or out of service, the call cannot be connected. Use of secondary routing increases the chances of successfully connecting a call, should the primary choice fail, while improving both the GoS offered to the customer and network resilience.

There are a number of alternative routing strategies available. The simplest alternative routing will attempt a second choice route if the first choice is busy. The problem with static alternative routing mechanisms is that they do not adapt themselves to changes in traffic loadings. A number of dynamic alternative routing schemes exist that help overcome this problem. Dynamic alternative routing with random routing allows alternative routes to be randomly selected from a list of those available. Dynamic alternative routing with blocking feedback routing overflows to a currently nominated path. If this route is busy, the call connection fails but a new nominated path is identified for the next overflow call. Dynamic alternative routing with least-loaded path routing uses an alternative

path with the highest availability. Primary traffic is protected by ensuring selection is made only if the number of idle links exceed a pre-set threshold.

As technology moves forward, switch processing and signalling capabilities become more powerful. This opens up the opportunity to route intelligently using a real time database that can detail the busy/idle state of the links. By monitoring the complete network, routing schemes can be adapted or re-arranged to connect calls with the minimum number of links. Each time a call clears, the freed links are examined to see if they can replace existing tandem routes if a shorter more direct link can be established.

5. Network growth design

The initial greenfield design for a new mobile network will be based on capacity estimates for the first year or two of operation. This is essential to ensure the cost of installing the new network is not prohibitive; however, the operator must design into the new network the capability for expansion as the demand for their network and services grows. The expansion of the network needs to occur as a whole, there is little point adding cell sites to relieve radio congestion if the fixed network does not have the capacity to support the additional traffic, and vice versa. Network growth must be carefully planned. Some parts of the fixed network are expensive and will give marked step changes in overall capacity, such as the requirements for additional MSCs. Careful growth design and optimisation of resources may be able to save such large purchases.

5.1 Radio growth design techniques and tools

As the network matures and also as the customers become more reliant on the services offered by the mobile network, the traffic carried by the network will increase. Also the expected coverage and quality of service expected by the customer increases. While the services are new, customers are prepared to suffer the occasional dropped call; however, as the network grows the expectation of quality coverage also grows. The radio network designer, therefore, has to expand not only the capacity of the radio network, but also the quality of coverage. This may call for specialist techniques such as cell enhancers to cover tunnels or microcells to provide exceptional capacity relief in large cities.

5.1.1 Initial capacity expansion

It is likely the first limitation that a greenfield network will experience is that of capacity within towns and cities. The green field network would have been designed to provide mainly coverage, rather than capacity, but as the traffic grows the limiting factor on the GoS will change from coverage to capacity issues. Experience has shown

that initial traffic levels grow quickly in urban areas initially and then even out to linear growth over the whole network.

The limiting factor to capacity is spectrum, the most valuable resource of any mobile operator. In order to improve capacity over the radio interface, spectrum is required; this can be achieved in several ways, ranging from improving frequency reuse by cell splitting, utilising system features to improve capacity performance or even applying to the local regulatory authority for additional spectrum.

The simplest approach is to add carriers to existing cell sites. However, this assumes that additional spectrum is available, or the initial frequency reuse of the roll-out network is well within the operating limits of the cellular system. The disadvantage of this approach is that it cannot extend the coverage of the network.

Another approach is to sectorise omni-directional cells. This instantly increases the capacity and improves the spectrum efficiency. In omni-directional cells, interference comes from all directions; however, the use of directional antennas limits the direction of transmission and hence the spread of interference. This allows for a more rapid frequency reuse, thus increasing capacity for a given spectrum allocation. The effect of sectorisation is shown in Fig 10. The coverage of the original omni-directional cell is shown by the solid hexagons and the effect of sectorisation is shown by the dotted lines radiating from the centre of each cell. Each original cell becomes three cells, each covering one third of the original cell's area; this has effectively increased the network capacity by a factor of three, assuming the same number of carriers can be assigned to each sector as the original omni-directional cell was allocated.

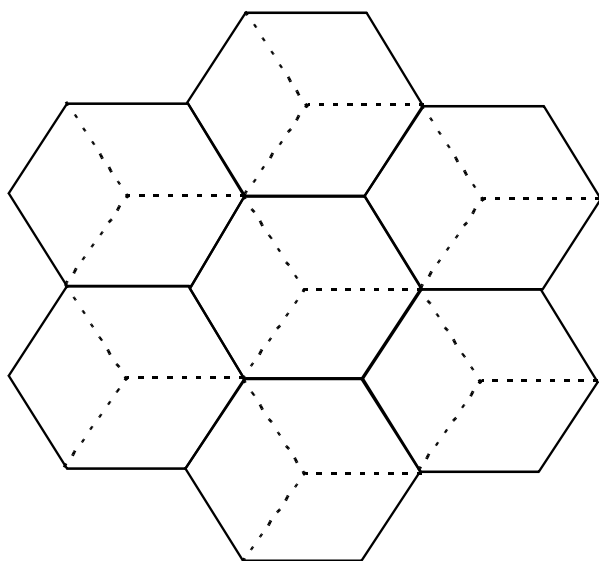


Fig 10 Example of cell sectoring.

Another approach in the initial stage of capacity growth is to cell split; this is a technique where additional base sites are added to the cellular network at places where coverage is weakest, such as at the edge of the coverage of each existing cell as shown in Fig 11. Although, in practice, the cell coverage area would not be an ideal hexagon but a non-uniform shape determined by local terrain features, hexagons are shown to demonstrate the point. Here the existing roll-out cells are shown as the solid hexagons and the dotted hexagons show where new cell sites could be installed to perform cell splitting. This layout effectively reduced the size of each of the existing cells, thus reducing the traffic loading and hence congestion. Figure 11 shows that the centre original cell would be left to service only a small proportion of its original coverage area; this has the advantage of both increasing capacity and improving coverage. The coverage is increased because the new cells will provide strong signal strengths in areas which were originally poorly served by the original cells. The approach has not only relieved the congestion for the centre cell, but also improved the coverage and hence GoS of the network as a whole in that area. The process of cell splitting can be applied until cell radii of only 500 metres or so are obtained. Beyond that frequency, planning becomes almost impossible and specialist techniques or microcells become essential.

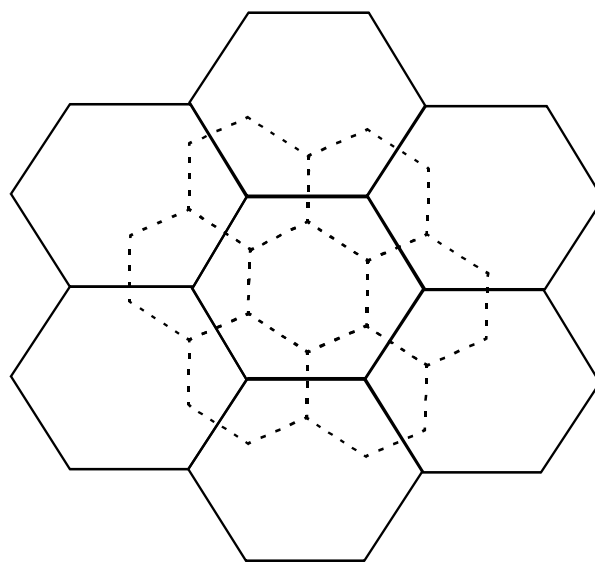


Figure 11 Example of cell splitting.

Other ways of improving capacity, assuming coverage is sufficient, involve increased spectrum efficiency. One of the biggest problems in the radio environment comes from the rapid changes in signal strength which can be experienced, for example, moving behind a large building and shadowing the mobile terminal from its cell site, or multi-path propagation which generates standing wave patterns around the mobile with small pockets where virtually no signal can be received due to near-perfect cancellation of two different paths half a wavelength apart.

To ensure that acceptable coverage and quality is maintained in these situations the radio planner allows large margins (up to 10 dB) in the design performance limits. Such margins can severely limit the overall capacity and ways of reducing these margins can improve the overall network capacity without the need for expensive cell splitting.

One of the simplest ways to reduce the effects of multi-path propagation is to use diversity; this can take the form of either space or frequency diversity. Space diversity employs multiple antennas, several wavelengths apart, which receive the transmitted signal and combine them in such a way as to avoid one antenna suffering a null. This is very effective, but the provision of dual antennas and the combining hardware prevents its use within hand held mobile equipment. Space diversity therefore, is limited to use on the uplink from the mobile to the cell site. Another method of applying diversity is to repeatedly and rapidly change frequency. The standing waves are frequency-dependent and a small change in frequency can move the mobile out of a null. The GSM system has the ability to perform frequency hopping using a feature called slow frequency hopping (SFH), where each time-slot is transmitted on a different carrier frequency. The mobile needs to know the hopping sequence so it can receive and transmit on the correct frequency for each time-slot and this is initiated by a complex synchronisation at call set-up. Diversity techniques vary in efficiency but generally a reduction in planning margin of about 2 dB can be achieved [1].

Other ways to improve spectrum efficiency rely on the statistical nature of calls on the system to reduce interference levels, thus improving call quality. These include features such as power control and discontinuous transmission (DTX). The GSM system supports both these features. Power control allows the transmit powers of both the mobile unit and the cell site to be reduced where strong received levels are obtained, such as when the mobile is close to the cell site. In this situation the transmit powers can be reduced while maintaining call quality. This may significantly help a mobile operating on a different co-channel cell site by reducing interference. DTX helps interference in a similar way, in that time-slots are only transmitted if there is information to be sent. As speech is typically present for only 50% of a conversation in any direction, this can halve interference levels. The effect of these methods combined with diversity techniques can significantly reduce planning margins from the original 10 dB, to perhaps 4 or 5 dB.

The GSM system also offers a further method to improve capacity, the half-rate speech codec. This is a reduced bit rate voice codec which occupies only half of a GSM time-slot. Thus, if two mobiles are using the half-rate codec, two calls can exist on one time slot, doubling the

capacity of the system; however, this assumes that all mobiles are capable of using the half-rate codec and that it gives acceptable quality. Early GSM mobiles use the standard full-rate codec and half-rate codecs are not currently being supported by most manufacturers because of the quality it exhibits. Enhanced full- and half-rate codecs are being developed [2] which give better quality in interference limited conditions, but the radio designer cannot make use of these gains until the majority of mobiles in the network are capable of supporting the new technology. Current mobile unit replacement rates would suggest that networks could not be re-dimensioned to make use of improved codec technology for at least three years after mobiles utilising the feature start appearing on the market.

5.1.2 Specialist coverage

As the network matures, certain features may cause performance limitations on the network. Typical problem features are road and rail tunnels, long deep mountain valleys, desert roads, etc. As the customer base expands, coverage of these features becomes important to the operator, and the radio planner has to find cost-effective but efficient methods to obtain quality service in these areas.

Tunnels are generally best covered by cell enhancers. These are basically two highly directional antennas connected back-to-back. One antenna is directed into the tunnels while the other is directed at the closest cell site. The effect of the two high-gain antennas is to provide sufficient signal into the tunnel to provide acceptable coverage. The tunnel walls also help to duct the signal along its length. Very long tunnels may not be fully covered by cell enhancers and may require the installation of cables known as 'leaky feeders'. This is effectively coaxial cable with a poor quality outer sheath allowing controlled loss of the signal along the cable. This couples well to antennas close by, such as a vehicle-mounted antenna within a tunnel. The leaky feeder can either be driven by a dedicated cell, or may repeat off-air signals derived from a local cell site.

The problem of cost-effective coverage to roads and railways in large unpopulated areas is best solved using highly directional antennas. The high gain antenna can increase the coverage area of a cell extensively, but only in one direction. Features like this can therefore be covered by a two-sectored cell with a highly directional antenna pointing along the road or railway. The number of sites required to achieve full coverage is reduced because the high-gain antennas allow for larger separation between sites compared with that for conventional omni-directional cells which would also provide coverage to all the surrounding areas.

5.1.3 Microcells

In a mature network, cell sizes may start to become very small following progressive cell splitting. The radio network designer has then to resort to the use of microcells to achieve the required capacity. Microcells differ from conventional macrocells in that their antennas are generally below roof level and thus the radio signal is, to some extent, contained by the surrounding buildings. Microcells offer high capacity to small confined areas. Typically such cells have a coverage area of about 200 metres. Therefore careful siting of microcells is paramount. Initially detailed site surveys need to be performed to ensure the microcell will support sufficient traffic. Tools are beginning to be developed which will aid the planner in this process by identifying likely traffic hot-spots. Intuition also suggests places such as main-line railway stations may be good candidates for such technology.

Microcells are very good at providing in-building coverage. The outer skin of a building usually provides quite-high attenuation, limiting the extent to which macrocells can provide in-building coverage, especially on lower floors. The use of a microcell in this situation can provide high-quality coverage with limited interference to the street level outside which can still be covered by conventional macrocells.

5.1.4 Radio planning tool requirements

Radio planning tools are lucrative business at the current time with many mobile operator licences on offer throughout the world. An accurate radio planning tool is essential for both the greenfield radio design as well as the design of capacity enhancements. The radio planning tool is the only realistic way to calculate predicted signal strength coverage and interference levels and hence produce a radio network design which meets the overall GoS.

The radio planning tool comprises three main parts, a database which contains the cell site locations together with their configurations, a radio propagation prediction engine to produce the signal strengths and interference levels, and an automatic frequency planning (AFP) module. The prediction engine for macrocells is normally an implementation of one of the many generalised radio propagation models such as Okumura [3] or Hata [4]. However, radio propagation from microcell antennas mounted below roof level is somewhat different to propagation from macrocells where the antennas are mounted in clear space. Different propagation models have been developed to give estimates of the propagation coverage, but the real challenge to the radio planner is to predict interference levels. Streets of tall buildings are very good at ducting radio waves, so accurate propagation predictions are increasingly utilising building databases and small-scale mapping data to give road orientation

information which can then be utilised by the propagation models. Models are currently being developed which perform full 3-dimensional ray-tracing of all possible radio paths between the microcells and any surrounding point [5]. Such models are in their infancy and, at the moment, need large amounts of computer time to run; however, with the ever-increasing power of computers, they may soon be feasible for incorporation into general radio planning tools.

The allocation of frequency channels to cells in a cellular network is a critical element of the design process, it affects the two major metrics of any cellular network — capacity and quality of service. AFP processes can efficiently and quickly generate a near-optimum frequency plan requiring only minor operator intervention. The AFP process calculates the potential interference between all cells in the network and forms a list of cells which cannot share the same frequency without infringing the GoS goals of section 2. Channels can then be assigned to each cell which will provide an acceptable GoS.

BTL has developed a fully functional radio planning tool called 'Saturn'. It is capable of providing the radio planner with all the detail necessary to successfully plan and implement a cellular radio network. It has been used to provide fully planned solutions to support bids for new mobile operator licences for a consortium of which BT is a partner.

5.2 Network growth design techniques and tools

Detailed statistics are available for the cellular network, reporting on cells, links and switches. Suitable interpretation of these statistics can give vital information on current performance, identifying congested or under-utilised resources. Extrapolation techniques can be employed to estimate likely future demand.

However, changes in customer behaviour due to regional variances, new services or new tariff structures can impact on the network. Such changes are slow to reveal themselves from analysis of statistics. Customer call records offer an alternative wealth of information on customer behaviour and calling patterns and can supplement the information provided by the statistics.

Suitable post processing of these call records can produce a traffic matrix reflecting the source-to-sink behaviour patterns of customers on the network. By applying this matrix to a modelling tool the planner is able to test the effect different customer profiles have on the network.

This section describes the modelling tool called GII used by BTL to model a cellular network. It was initially designed to model the current UK analogue cellular

network and is capable of representing the switches, links and cells. It offers the user control over cell parenting, topology, routing and traffic levels. Information on customer behaviour, provided via comprehensive traffic matrices, can be modified to reflect changes in calling patterns.

The tool is constantly being updated to reflect the live network and will soon be capable of representing the hybrid networks. Matrices built from both digital and analogue system call records will be needed to represent the behaviour of all customers correctly. Digital networks are expected to have a different usage profile, especially with the increased use of data calls rather than just voice traffic. The model is a powerful 'what if' tool capable of showing the impact customer demand has on the network. Switch and link loadings are reported and inter-switch links dimensioned. Estimated revenue figures are given to allow different routing and interconnect strategies to be evaluated. The signalling loads are computed and the impact of cell parenting on hand-off and paging loads are reflected.

A graphical interface (GIS) for GII is under development, which can present results quickly and show whether a design is sensible. Hot spots, under-utilisation or inefficient routing can be readily identified. The whole network will be displayed in skeleton form, collapsed to show the traffic flows between nodes. Zooming functions will be available to expose more detail on a particular part of the network. Drop-down menus and supplementary data boxes will be displayed by clicking on an area of interest within the network. Once the basic design has been reviewed hard copies of both tabular and graphical reports can be generated.

6. Conclusions

This paper has followed the birth and growth of a cellular network and shown how BTL has met the challenges that this presents. BT's experiences have come both from working with Cellnet in the UK and in overseas bids and network operations. Cellular has now reached its adolescence and is entering adulthood. Already cellular growth, with over 2 million new cellular connections in 1995 in the UK, exceeds fixed network growth and with this equating to an annual rate of about 70% it may not be too long before the number of cellular connections exceeds that for the fixed network.

References

- 1 Zako B, Ibbetson L and Lopes L: 'The GSM radio link performance with space diversity and slow frequency hopping', 43rd IEEE Vehicular Technology Conference Personal Communication — Freedom Through Wireless Technology (Cat. No.93CH3305-0), pp 480—482 (May 1993).
- 2 Wong W T K, Mack R M, Cheetham B M G and Sun Z Q: 'Low rate speech coding for telecommunications', BT Technol J, 14, No 1, pp 28 — 44 (January 1996).
- 3 Okumura Y et al: 'Field strength and its variability in VHF and UHF land mobile radio services', Rev of the ECL, 16, pp 825 — 873 (1968).
- 4 Hata M: 'Empirical formula for propagation loss in land mobile radio services', IEEE Trans, VT-29, pp 317—325 (1980).
- 5 Sharples P A and Mehler M J: 'Propagation modelling in microcellular environments', IEE Conference Publication 370 (ICAP93), pp 68—71, (1993).



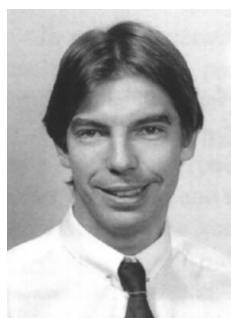
John Button joined BT Laboratories in 1979 as an apprentice. In 1982 he joined the radio communications division where he was involved in the development of cordless telephones. He obtained his BSc in Telecommunications from the University of Essex in 1987.

Since graduating he has worked for BTL on radio planning aspects of mobile communications including digital cordless communications systems and conventional analogue and digital cellular communications systems.



Ken Calderhead graduated from Strathclyde University in 1979 with an honours degree in applied physics. Following post graduate research in environmental physics he joined the teletraffic division at BT Laboratories in 1983. He has been actively involved in cellular network performance, quality and design issues since 1987 and is currently in the cellular systems unit.

He is a member of the British Computer Society and is a Chartered Engineer.



Ian Goetz joined BT Laboratories in 1987 in the transmission performance unit, working in a team to develop a new type approvals tester for ISDN telephones and a real time DSP system to model network speech paths for subjective and objective assessment. In 1990 he joined the cellular systems unit working on mobile transmission performance and transmission design issues. In this role he has represented BT on GSM, DECT and CT2 standards bodies and worked closely with Cellnet on transmission network design issues. In addition, he designed the transmission and interconnect for the

successful BT bid for the second GSM licence in Spain. He is now working with Cellnet.



Richard Patient joined BT Laboratories as a technician in 1984. He gained a BSc in electronic engineering from the University of Kent in 1991, and after graduating joined the cellular systems unit. He has worked on the use of automated cellular network call quality measurement systems, including the Oftel survey of the Vodafone and Cellnet networks. He is currently providing support to Cellnet in planning and implementing the migration of spectrum from the TACS to the GSM network, specifically looking at techniques to maintain capacity and call quality. He also represents BT on the ETSI Sub-Technical

Committee involved with GSM radio standards.



Martin Hodges joined BT in 1979. He graduated from Essex University in 1976 with a degree in telecommunications engineering and gained his MSc and PhD in telecommunications systems from there in 1977 and 1979 respectively. He joined BT Laboratories to work on digital speech interpolation systems and worked on the evaluation of systems for TAT8 with particular emphasis on the variable rate speech coding element. His involvement in speech coding grew, acting as technical consultant to the CCITT group standardising G.721 (the 32 kbit/s ADPCM coder) and then

helping to develop the BTL codec used in Skyphone on which he holds a patent. In 1985 he moved to head a group on cellular radio and was actively involved in the early phases of the GSM standardisation process. He is currently leading a team, in the cellular systems unit, which supports the radio development of the Cellnet TACS and GSM networks.



Richard Pilkington joined BT Laboratories with a first class honours degree in applied physics from Portsmouth Polytechnic in 1986 and was involved in the specification and testing of the first GSM system and the radio interface for this. In 1994 he took over the development of analysis tools for the cellular networks and has led the development of the versions of GIANT delivered into Cellnet.

His current role also includes leading the development of the Auto site placement tool.



Ray Reeve joined BT in 1966 as an apprentice. He went on to head a team of engineers on Exchange maintenance and gained a City and Guilds and HNC in electrical engineering. He later joined the performance engineering division to work on PABX type approval, TXE4 dimensioning and on development work for the Monarch range of PABXs. He has supported Cellnet since their launch date, analysing statistical and call record information and developing an on-line statistics reporting package. He was responsible for the development of the GII network modelling tool and currently

maintains an interest in its further development to include GSM modelling.



Paul Tattersall has a background of working in a local telephone area, hardware and radio measurements on the trunk circular waveguide project and switching experience gained through the early development of the UXD5 small digital telephone exchange. He began work in cellular radio at the time the TACS air interface standard was being formed and was later involved in the development of the GSM radio interface standards through laboratory and field trials. His recent cellular work has mainly been directed at validating new GSM network features and designs, which strive to increase

capacity and improve quality, while holding costs down. This has necessitated a close understanding of cellular network design and development of quality of service measurement and evaluation methods. He currently leads a team within the cellular systems development unit at BT Laboratories.